Chapter 1

What is Numerical Analysis about ?

Case 1. Minimizer

Theorem 1. *Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$. Let $m := \inf\{f(x) \mid x \in [a, b]\}$. Then there exists a minimizer $x_* \in [a, b]$ such that $f(x_*) = m$.*

Note: we have used that a continuous function f on $[a, b]$ is bounded.

Proof. Step 1. For each $n = 1, 2, 3, \cdots$ let $x_n \in [a, b]$ be any element such that

$$f(x_n) < m + \frac{1}{n}.$$

For each n, such an element must exist: Otherwise, i.e., if $f(x) \geq m + \frac{1}{n}$ for every $x \in [a, b]$, then $m + \frac{1}{n}$ is a lower bound of the range set that is strictly bigger than m, contradicting to the definition of m.

Step 2. We have defined the sequence (x_n), and let $y_n := f(x_n)$. Then $y_n \to m$ as $n \to \infty$ because

$$0 \leq |y_n - m| < \frac{1}{n}.$$

Step 3. (x_n) is a sequence of real numbers in $[a, b]$, and thus (x_n) is a bounded sequence. Therefore, there exists a convergent subsequence $(x_{n_k})_{k=1}^{\infty}$ to a limit x_*. Furthermore, because $[a, b]$ is closed, the limit point $x_* \in [a, b]$. We also know that the subsequence (y_{n_k}) of a convergent sequence (y_n) also is convergent to the same limit m.

Step 4. Now, we observe

$$m = \lim_{k\to\infty} y_{n_k} = \lim_{k\to\infty} f(x_{n_k}) = f(x_*),$$

where in the last equality, we used the fact that f is continuous on $[a, b]$. $\square$

Case 2 : rank of a matrix

Let $A \in \mathbb{R}^{n\times k}$ be a real $n \times k$ matrix. The very first fundamental notion for A is the **rank of** A, the number being **defined** by the fact that

the number of linearly independent columns = the number of linearly independent rows.

Once the assertion of the equality is proved, the definition of the rank will make sense.

Denote the $(LHS) = r_{col}(A)$ and $(RHS) = r_{row}(A)$. This fundamental equality is a consequence of the inequality that

$$\text{for any matrix } A, \quad r_{row}(A) \le r_{col}(A)$$

because if that is true then $\Big(r_{col}(A) = \Big)r_{row}(A^T) \le r_{col}(A^T)\Big(= r_{row}(A)\Big)$.

The inequality follows from the assertion

if $r_{i_1}, r_{i_2}, \cdots r_{i_\rho}$ are linearly independent rows of A
then $Ar_{i_1}, Ar_{i_2}, \cdots Ar_{i_\rho}$ are linearly independent vectors in $\mathbb{R}^n$.

Here we denote the rows of A from top to bottom by r_1, r_2, $\cdots$, r_n. We examine the proof of the assertion.

Proof. Suppose not, i.e., we have $(\lambda_1, \lambda_2, \cdots, \lambda_\rho) \neq 0$ such that $\sum_{\alpha=1}^{\rho} \lambda_\alpha\big(Ar_{i_\alpha}\big) = 0$. Then,

$$0 = \sum_{i=\alpha}^{\rho} \lambda_\alpha (Ar_{i_\alpha}) = \sum_{\alpha=1}^{\rho} A(\lambda_\alpha r_{i_\alpha}) = A\Big(\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha}\Big).$$

But,

$$Ax = 0 \quad \Longleftrightarrow \quad x \perp \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle.$$

Combining the two observations,

$$\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha} \in \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle \cap \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle^{\perp} = \{0\}.$$

Thus, we conclude that $\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha}$ is a zero vector. Recalling that $r_{i_1}, r_{i_2}, \cdots r_{i_\rho}$ are linearly independent, we conclude $\lambda_1 = \lambda_2 = \cdots = \lambda_\rho = 0$. Contradiction. $\square$

This is the way we are taught in Linear Algebra course: we prove the equality, and based on this fact, we give the definition for the rank. However, this proof does not give a clue on how do we know the rank of A.

This second example is the case where, there exists another proof that contains a clue for the calculation.

Singular Value Decomposition

Theorem 2. *For any matrix $A \in \mathbb{R}^{n\times k}$, then there exist*

1. *an $n \times n$ orthogonal matrix O,*
2. *a $k \times k$ orthogonal matrix Q,*
3. *an $n\times k$ matrix Λ where nonzero entries are only at diagonals Λ_{ii}, $i = 1, 2, \cdots, min(n, k)$*

so that $A = O\Lambda Q^T$. The number of nonzero entries of Λ coincides with the rank of A.

I will not provide the proof of the Singular Value Decomposition, but its proof contains a way to compute O, Q, and Λ, and this provides the mean to compute the rank.

Therefore, in some cases, there can be two proofs:

1. one that is optimal for the proof, easy to read, maybe short, etc.
2. the other that is optimal for the implementaion, maybe longer, etc.

so the material for the Numerical Analysis may be taught from different point of view.

Case 3 : Riemann Integral or Quadrature

The third example is the case, where the definition contains a clue for a computation.

Let us consider a high-school series calculations. Let n be a fixed natural number. For each i, we let $x_i = a + i\frac{b-a}{n}$. We let t_i be any point belogs to $[x_{i-1}, x_i]$ for $i = 1, \cdots n$.

The definition of the Riemann Integral of a continuous function f over $[a, b]$ is justified by **the existence of a real number** I such that

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n} = I.$$

regardless of choice (t_i) at each n. We examine the proof of the assertion.

Proof. Step 1. By extreme value theorem on each compact interval $[x_{i-1}, x_i]$, there exist the minimizer t_{i*} and t_i^* in $[x_{i-1}, x_i]$. Then we have for each n

$$\sum_{i=1}^{n} f(t_{i*})\ \frac{b-a}{n} \le \sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n} \le \sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n}.$$

Step 2. We omit the proof that $\left(\sum_{i=1}^{n} f(t_{i*})\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ and $\left(\sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ are bounded monotone sequences so that their respective limits $\alpha, \beta \in \mathbb{R}$, $\alpha \le \beta$ exist.

Step 3. That f is continuous in $[a, b]$ implies the following:

$$\begin{aligned}&\text{For every } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that}\\ &x, x' \in [a, b], |x - x'| < \delta \quad \Longrightarrow \quad |f(x) - f(x')| < \frac{\epsilon}{b-a}.\end{aligned}$$

that is to say, f is uniformly continuous in $[a, b]$. For a given ϵ, we take $K \in \mathbb{N}$ so that $\frac{b-a}{K} < \delta$ for each δ. This gives that

for every ϵ, there exists K such that for $n \ge K$,

$$\Big|\sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n} - \sum_{i=1}^{n} f(t_{i_*})\ \frac{b-a}{n}\Big| \le \sum_{i=1}^{n} \Big|f(t_i^*) - f(t_{i*})\Big|\ \frac{b-a}{n} \le \sum_{i=1}^{n} \frac{\epsilon}{b-a}\ \frac{b-a}{n} = \epsilon.$$

This implies that the limits $\alpha = \beta$. By the Squeeze theorem, $\left(\sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ is convergent, to the same limit. The limit is defiend to be I. $\square$

Therefore in some cases,

1. the proof we are taught contains a well-working method.
2. Since there is no reason we stick to this method only, of course we explore quite much better working numerical methods (midpoint rule, simpson rule, gaussian quadrature, $\cdots$) than the one suggested.

Case 4 : Jordan factorization of a square matrix T.

Another striking example where we do not have a general algorithm stably working for it is the Jordan factorization.

Theorem 3. *Let A be an $n \times n$ real square matrix. Then there exist*

1. *an $n \times n$ invertible matrix P,*
2. *an $n \times n$ matrix J in real Jordan form,*

such that $A = PJP^{-1}$.

Try to re-visit its proof in Linear Algebra textbook. They only assert the existence. More precisely, the steps that possibly compute P and J contains the precedure whose small perturbation gives rise to completely different result.

What will we treat in this course ?

1. Definition of a sequence (a_n), and a function $f : [a, b] \to \mathbb{R}$.
2. interpolating / approximating polynomials for a class of functions. Topics: spline, Tchebyshev polynomials, $\cdots$.
3. Existence of fixed point and Newton's method on nonlinear system.
4. Existence of solution of initial value problem of nonlinear ode / Runge-Kutta methods, $\cdots$
5. For all of above, we will conduct the error analysis.
6. (2nd semester) Linear algebra theory or higher dimensional techniques and more on (partial) differential equations.

Chapter 2

How does my computer throw $\sin(x)$ for an input x?

Case 1. A function from $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ to $\mathbb{R}$

We can define a function $a : \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \mapsto \mathbb{R}$, using the table.

n	1	2	3	4	5	6	7	8	9	10
a_n										

Case 2. A function from $\mathbb{N}$ to $\mathbb{R}$

Are there ways to define a function from countably infinite set to $\mathbb{R}$?

Case 3. A function from $[0, 1] \subset \mathbb{R}$ to $\mathbb{R}$

Are there ways to define a function from this uncountably infinite set $[0, 1]$ to $\mathbb{R}$?

A function on $E \subset \mathbb{R}^n$?

Is a function a black box ?
Is a function a table ?
Can you come up with an example of a function that you can build as a machine ?
How does my computer know which value to return for $\sin(x)$?

A few observations

1. A recursive definition for a sequence
   - For a sequence (a_n) it is possible to have a recursive definition.
   - A recursive definition makes sense on the set $\mathbb{N}$, and in general on the set with *well-ordering.*
   - $[a, b]$ *with usual ordering* $<$ *is not the case.*

2. The use of exponential b^n
   - For a given number b, b^n with n integer can be computed by finite use of arithmetics.
   - This is again a privilege of sequence which is not given to the function:

     $$\text{Let} \quad f(x) = b^x, \quad x \in \mathbb{R},\ b > 0.$$

     in case x is irrational number, b^x cannot be done in the same way as before.

3. For the use of $id(x) : x \mapsto x$, one sees that
   - The return value is computed <u>on the fly</u> at the time x is decided.
   - Hence, what we are doing is not like making a table, but more resembles the *pinball* machine.
   - As its consequence, in reality this device will handle only the rational input number x.

It seems, there are not many others than the class of elementary pp functions.

I wanted you to see the importance of a class of functions on $\mathbb{R}^n$ that are

(1) piecewise polynomial, and

(2) each piece's domain (the support more precisely) is elementary in the sense that you can finish the membership check of a given x to the domain in a finite procedures.

We will call those functions an elementary pp function.

Examples of sets $E \subset \mathbb{R}^n$, where the membership check is finitely done.

Remark 2.1. The literal definition of the function on an uncountable set, which might suggest the tabularized mapping, indeed is beyond our intuition.

Remark 2.2. For functions that are continuous or measurable, the relation between elemetary pp functions and them are like the relation between rational numbers and real numbers.

A continuous function on $\mathbb{R}^n$ is **a limit of continuous elementary piecewise polynomial functions**, where the convergence is uniform in every compact subset of $\mathbb{R}^n$.

A measurable function on $\mathbb{R}^n$ is **a limit of elementary piecewise polynomial functions**, where the convergence is pointwise, for almost everywhere $x \in \mathbb{R}^n$.

Therefore, in principle, a suitable quantity can be computed as accurate as we want. This yields the concept of computing the limit and error control.

Objective of our study

1. We may assume sufficiently but finitely many sampling data for a function f:

$$\big(x_i, f(x_i)\big)_{i=1,\cdots,N}.$$

2. For this sample data, we come up with a pp function φ such that

$$\varphi(x_i) = f(x_i), \quad i = 1, 2, \cdots, N.$$

3. Out of
   (a) The sample data,
   (b) The assumption that f was, say m times continuously differentiable.

   we look for the remainder formula for other values of x

$$R(x) = f(x) - \varphi(x).$$

Questions

1. How do we choose a pp function φ? $\rightarrow$ theory and method
2. How accurate is it ? $\rightarrow$ The remainder formula.

One piece vs. manu pieces

For these two questions,

First, we will take one piece, out of piecewise polynomial function, and seek for the remainder formular, restricted on that piece's support. (Local study)

Second, we assemble pieces into one pp function. (Global study)

One piece: (Local) polynomial interpolation

Let $I = [a, b]$. Fix an integer $n \geq 0$.

Sample Data

With repect to the integer $n \geq 0$, we choose, or we are given, the $n+1$ sampling points

$$x_1, x_2, \cdots, x_{n+1} \in [a, b].$$

The choice then will assume in our class:

$$\text{If } x_i \text{ is repeated } m \text{ times} \quad f(x_i), f'(x_i), \cdots, f^{(m)}(x_i) \quad \text{are collected at } x_i,$$

where f is sufficiently many times differentiable.

Sets of functions

We consider

$$\begin{aligned} C^{n+1}([a,b]) = \big\{ f : [a,b] \to \mathbb{R} \mid f \text{ is } n+1 \text{ times differentiable and} \\ f,\ f',\ f'',\ \cdots,\ f^{(n+1)} \text{ are continuous in } [a,b] \big\}. \end{aligned}$$

Note: At the endpoint a and b, the differentibility is the right-differentiability and the left-differentiability respectively.

We also let

$$\mathcal{P}^{<n+1}([a,b]) = \text{the set of polynomials of order at most } n \text{ restricted on } [a,b].$$

Interpolation problem

For a given choice,

1. We look for a polynomial $p \in \mathcal{P}^{<n+1}([a,b])$ that matches the data, for example,

$$p(x_i) = f(x_i), \quad p'(x_i) = f'(x_i), \quad \cdots, \quad p^{(m)}(x_i) = f^{(m)}(x_i).$$

2. We seek for a formula of the remainder $R(x)$,

$$R(x) = f(x) - p(x), \quad x \in [a,b]$$

In fact, the term *interpolation* does not necessarily assume the function f and does not assume that the data are collected by sampling. Nevertheless, it is convenient to speak of the function f.

For those who major in this field, the classes of functions we can treat similarly are

$$\begin{aligned} C^{n+1}([a,b]) \subset O^{n+1}([a,b]) &= \big\{ f : [a,b] \to \mathbb{R} \mid (n+1)\text{-th distributional derivative of } f \text{ is in } L^1([a,b]) \big\} \\ \subset \tilde{O}^{n+1}([a,b]) &= \big\{ f : [a,b] \to \mathbb{R} \mid (n+1)\text{-th distributional derivative of } f \text{ is a Radon measure.} \big\} \end{aligned}$$

Fundamental Theorem of Calculus, Taylor's Theorem

We recall first the fundamental theorem of calculus:

Theorem 1. *Let $f \in C^1([a,b])$ and $x_1 \in [a,b]$. Then for $x \in [a,b]$,*

$$R(x) = f(x) - f(x_1) = \int_{x_1}^{x} \frac{df}{dx}(x')\, dx' = (x - x_1) \int_{\tau=0}^{1} \frac{df}{dx}\big((1-\tau)x_1 + \tau x\big)\, d\tau.$$

Mean Value Theorem: there exists a $\xi \in [a,b]$ such that

$$R(x) = f(x) - f(x_1) = (x - x_1) f'(\xi).$$

Theorem 2 (Taylor's theorem)**.** *Let $f \in C^{n+1}([a,b])$ and $x_1 \in [a,b]$. Then for $x \in [a,b]$,*

$$\begin{aligned} R(x) &= f(x) - \Big(f(x_1) + f'(x_1)(x - x_1) + \frac{f''(x_1)}{2!}(x - x_1)^2 + \cdots + \frac{f^{(n)}(x_1)}{n!}(x - x_1)^n\Big) \\ &= (x - x_1)^{n+1} \times \frac{1}{n!} \int_{\tau=0}^{1} (1-\tau)^n \frac{d^{n+1} f}{dx^{n+1}}\Big((1-\tau)x_1 + \tau x\Big)\, d\tau. \end{aligned}$$

Mean Value like Theorem: there exists a $\xi \in [a,b]$ such that

$$R(x) = \frac{(x - x_1)^{n+1}}{(n+1)!} f^{(n+1)}(\xi).$$

Remark 2.3. We take the FTC as the $n = 0$ case of the Taylor's theorem. We take the Taylor's therem as the case where the choice of points and data were all at x_1,

$$x_1, x_1, x_1, \cdots, x_1 \quad (n+1) \text{ times.}$$

We will prove the similar remainder theorem for the arbitrary choice of the $n+1$ points.

Chapter 3

The notation $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$

Let $I = [a, b]$, $n \geq 0$ an integer, $\underline{f \in C^{n+1}([a, b])}$. Let $0 \leq \ell \leq n+1$ be an integer and we consider $\ell + 1$ points

$$x_1, x_2, x_3, \cdots, x_{\ell+1} \in [a, b] \quad \text{(may be repeated)}.$$

We will denote the choice of $\ell + 1$ points by

$$[x_1, x_2, \cdots, x_{\ell+1}].$$

A choice $[x_1, \cdots, x_{\ell+1}]$ acts on the class $\underline{C^{n+1}([a, b])}$, denoted by

$$\big([x_1, x_2, \cdots, x_{\ell+1}], f\big) \in \mathbb{R}.$$

For the sake of mathematical clarity, this action is defined by nontrivial surface integrals in the next page. Its motivational uses are given first.

Properties of the action $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$

(1) The value is independent of orders in the choice $[x_1, x_2, \cdots, x_{\ell+1}]$.

(2) Example of $\ell = 1$ case:

$$\begin{aligned} &\text{if } x_1 \neq x_2, && \big([x_1, x_2], f\big) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1) - f(x_2)}{x_1 - x_2}, \\ &\text{if } x_1 = x_2 = \bar{x}, && \big([\bar{x}, \bar{x}], f\big) = f'(\bar{x}). \end{aligned}$$

(3) For $1 \leq \ell \leq n+1$, the equality always holds:

$$\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)\ (x_{\ell+1} - x_1) = \big([x_2, x_3, \cdots, x_{\ell+1}], f\big) - \big([x_1, x_2, \cdots, x_\ell], f\big).$$

Indeed, above $\ell = 1$ calculations verify

$$\big([x_1, x_2], f\big)(x_2 - x_1) = \big([x_2], f\big) - \big([x_1], f\big).$$

Definition

Definition 1. *For $f \in C^{n+1}([a,b])$, and $0 \le \ell \le n+1$ integer,*

$$\begin{aligned}
&\text{if } \ell = 0 && \big([x_1], f\big) = f(x_1), \\
&\text{if } \ell \ge 1 && \big([x_1, x_2, \cdots, x_{\ell+1}], f\big) := \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell \subset \mathbb{R}^{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots \lambda_{\ell+1} x_{\ell+1}\big)\, dS(\lambda).
\end{aligned}$$

The set Λ^ℓ is the regular ℓ-simplex

$$= \Big\{(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}) \in \mathbb{R}^{\ell+1} \mid \forall e \quad \lambda_e \ge 1, \quad \sum_{e=1}^{\ell+1} \lambda_e = 1\Big\} \subset \mathbb{R}^{\ell+1}$$

The integral is denoted by $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$, or $[x_1, \cdots, x_{\ell+1}]f$, or $f[x_1, x_2, \cdots, x_{\ell+1}]$.

1d remainder theorem

Having defined the actions for $f \in C^{n+1}([a,b])$, we can state the remainder theorem.

Theorem (1d remainder theorem)**.** *Let $f \in C^{n+1}([a,b])$ and x_1, x_2, $\cdots$, $x_{n+1} \in [a,b]$. Then, for $x \in [a,b]$,*

$$\begin{aligned}
R(x) = f(x) - \Big(&\big([x_1], f\big) + \big([x_1, x_2], f\big)(x - x_1) + \big([x_1, x_2, x_3], f\big)(x - x_1)(x - x_2) + \cdots \\
&+ \big([x_1, x_2, \cdots, x_{n+1}], f\big)(x - x_1)(x - x_2)(x - x_3)\cdots(x - x_n)\Big) \\
&= (x - x_1)(x - x_2)\cdots(x - x_n)(x - x_{n+1})\big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big).
\end{aligned}$$

Remark 3.1. .

1. Note that the remainder is in the form of integral, that is the definition $\big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big)$.
2. We will also give its *mean value evaluation* too.

$$\exists \xi \in [a,b] \quad \big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big) = \frac{1}{(n+1)!} f^{(n+1)}(\xi).$$

3. Taylor's theorem is a special case where $x_1 = x_2 = \cdots = x_{n+1}$.

Proof of 1d remainder theorem

All we need is the key equality

$$f[x_1, x_2, \cdots, x_{\ell+1}]\ (x_{\ell+1} - x_1) = f[x_2, x_3, \cdots, x_{\ell+1}] - f[x_1, x_2, \cdots, x_\ell]. \qquad \text{(KEY)}$$

We will give a proof of this Key equality, and mean value evaluation as well, later using the surface integral definition.

Proof. For for $k \in \{0, 1, 2, \cdots, n\}$, we let

$$\begin{aligned} p_{<k+1}(x) = \Big(&\big([x_1], f\big) + \big([x_1, x_2], f\big)(x - x_1) + \big([x_1, x_2, x_3], f\big)(x - x_1)(x - x_2) + \cdots \\ &+ \big([x_1, x_2, \cdots, x_{k+1}], f\big)(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k)\Big). \end{aligned}$$

We assert that for $k \in \{0, 1, 2, \cdots, n\}$,

$$f(x) - p_{<k+1}(x) = \big([x_1, x_2, \cdots, x_{k+1}, x], f\big)(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_{k+1}).$$

At $k = 0$, by the key equality

$$f(x) - p_{<1}(x) = f(x) - f(x_1) = \big([x_1, x], f\big)(x - x_1).$$

Now, if the assertion is true for $0, 1, 2, \cdots, k - 1$,

$$\begin{aligned} f(x) - p_{<k+1}(x) &= f(x) - p_{<k}(x) - f[x_1, x_2, \cdots, x_{k+1}](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &= f[x_1, x_2, \cdots, x_k, x](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &\quad - f[x_1, x_2, \cdots, x_k, x_{k+1}](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &= f[x_1, x_2, \cdots, x_k, x_{k+1}, x]\Big((x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{k-1})(x - x_{k+1})\Big). \end{aligned}$$

□